EXPERIMENT - 9

IMPLEMentation of Subnetting in cisco packet Tracker simulation.

AIM:

To implement subnetting in cisco packet Tracker by creating multiple subnets, assigning static IP address connecting the subnets using routes testing Packet transmission

Requirements:

cisco packet tracker.

Router

switches

PCs

Copper straight - through cables.

Procedures:

1) Open cisco packet Tracer and create a new network topology

2) Add the required routers, switches and PCs to the workspace.

3) Connect the PCs to switch to the router using suitable cables.

4) Divide the network 192.168.1.0/24 into smaller subnet using the 127 subnet mask.

6) Configure the router interface through the CLI

Example:

enable
Configure terminal
IP address 192.168.1.1. 255.255.255.224
no shutdown
exit
interface FastEthernet0/1
IP address 192.168.2.1.255.255.255.224
no shutdown
exit.

7) Configure IP address, subnet mask & default gateway on each PC

8) check the connection b/w PCs using Ping Command.

10) Observe packet transmission with and without router.

Device	IP address	subnet mask	Gateway
Router	192.168.1.1	255.255.255.224	—
PC1	192.168.1.11	255.255.255.224	192.168.1.1
PC2	192.168.1.12	255.255.255.224	192.168.1.1
PC3	192.168.1.13	255.255.255.224	192.168.1.1

Result:

Thus subnetting was successfully implemented in Cisco packet Tracer.